This article has been accepted for publication in IEEE Transactions on Automatic Control. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TAC.2025.3647564

Distributed Zonotopic Fusion Estimation for Multi-sensor Systems

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***Abstract*— Fusion estimation is widely applied in multi-sensor systems to provide accurate state information, which is crucial for designing efficient control and decision-making strategies. Despite their ability to accommodate unknown noise statistics, set-membership approaches to fusion estimation are still in an early stage of development, especially zonotopic fusion estimation, which is highly advantageous given its computational efficiency and superior representational granularity. This paper is concerned with the distributed zonotopic fusion estimation problem for multi-sensor systems. The objective is to propose a zonotopic fusion estimation approach based on several zonotope fusion criteria. We first propose a novel zonotope fusion criterion to compute a distributed zonotopic fusion estimate (DZFE). The DZFE is formulated as a zonotope enclosure for the intersection of local zonotopic estimates from individual sensors. Then, the optimal parameter matrices for tuning the DZFE are determined through the analytical solution of an optimization problem. To reduce the conservatism of the DZFE with optimal parameters, we introduce an improved zonotope fusion criterion, which further improves the estimation performance by constructing tight strips for the intersection. In addition, we tackle the problem of handling sequentially arrived local estimates in realistic communication environments by introducing a sequential zonotope fusion criterion. This sequential zonotope fusion offers reduced computational complexity compared to batch zonotope fusion. The proposed zonotope fusion criteria are designed to meet the state inclusion property and to achieve superior performance compared with local zonotopic estimates. We also derive stability conditions for these DZFEs to ensure their generator matrices are ultimately bounded. Finally, two illustrative examples are employed to demonstrate the effectiveness and advantages of the proposed methods.**

***Index Terms*— Multi-sensor system, zonotope fusion criterion, sequential zonotope fusion, state inclusion property, performance superiority.**

This work was supported in part by the Joint Funds of the National Natural Science Foundation of China under Grant U24A20258, in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LRG25F030001, in part by the funding of Leading Innovative and Entrepreneur Team Introduction Program of Zhejiang under Grant 2023R01006, and in part by the Fundamental Research Funds for the Provincial Universities of Zhejiang under Grant RF-C2023007. (* Corresponding author)

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I. Introduction

A. Background and Motivation

In recent decades, multi-sensor fusion estimation has attracted considerable research interest due to its broad applications, including target tracking [1]–[3], localization [4], [5], environmental monitoring [6], [7], and health monitoring [8], [9]. The primary objective of fusion estimation is to improve state estimation accuracy by integrating valuable information from multiple sensors. For example, raw measurements from individual sensors can be transmitted to a fusion center (FC) for either measurement fusion [10]–[12] or centralized fusion estimation [13]–[15]. Alternatively, local estimators can be deployed at sensors to process raw measurements, after which the FC aggregates the locally computed estimates for distributed fusion estimation [16]–[18] Distributed fusion offers enhanced robustness, flexibility, and reliability compared to centralized fusion, owing to its parallel structure [18]. Moreover, local fault detection and isolation mechanisms in distributed systems mitigate performance degradation in the event of sensor failures. A key challenge in distributed fusion estimation lies in designing effective fusion criteria to consistently combine local sensor estimates while maximizing estimation accuracy. Existing fusion criteria predominantly assume Gaussian posterior estimates, yet real-world applications frequently encounter non-Gaussian noises with unknown statistical characteristics [19], [20]. Unfortunately, pre-identifying these noise distribution parameters through sensor system testing is often impractical. Set-membership estimation overcomes these limitations by bounding noise-admissible values into geometric sets [21], including ellipsoids, intervals, parallelotopes, zonotopes, and general polytopes. A comprehensive set comparison [22] shows that zonotopes achieve an advantageous balance between representation granularity and computational efficiency, specifically due to the following two characteristics: 1) Unlike coarse representations such as ellipsoids and intervals, a zonotope's granularity can be improved by increasing the number of generators; 2) In contrast to general polytopes, zonotopes enable exact and efficient closed-form computation of affine transformations and Minkowski sums, involving only simple matrix and vector arithmetic. These advantages have significantly facilitated the design of zonotopic estimators and are also beneficial for applications in multi-sensor fusion systems. However, designing zonotope fusion criteria is inherently difficult due to the infinite possibilities for seeking a zonotope enclosure of an irregular set.

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Based on the above considerations, we focus on the design of zonotope fusion criteria for distributed zonotopic fusion estimation.

B. Contributions

The following contributions are made in this paper:

1) **Zonotope Fusion Criteria:** We propose a zonotope fusion criterion for merging local zonotopic estimates into a distributed zonotopic fusion estimate (DZFE). The optimal parameter matrices for this criterion are obtained through the analytical solution of an optimization problem. Thereafter, we introduce an improved zonotope fusion criterion by constructing tight strips for the intersection of local zonotopic estimates to enhance the fusion performance. Compared with the weighted fusion [23], the proposed fusion criteria ensure the state inclusion property that guarantees the real state is always within fusion estimates. Different from our previous work [24], the proposed DZFEs have performance superiority over local zonotopic estimates. In other words, DZFEs that integrate information from multiple sensors provide better estimation performance than local estimates using a single sensor.
2) **Sequential Zonotope Fusion:** To reduce the computational burden, we propose a sequential zonotope fusion criterion that fuses local zonotopic estimates sequentially. Similar to the sequential fusion process in [25], the proposed sequential fusion criterion distributes the computational cost of the sequential DZFE over the entire time period by splitting the high-dimensional batch fusion process into multiple, low-dimensional fusion steps. Furthermore, the sequential DZFE maintains the state inclusion property and the performance superiority in the batch fusion criteria.
3) **Stability Analysis:** We conduct a stability analysis of the proposed DZFEs based on their performance superiority. The results indicate that the DZFEs are ultimately bounded if at least one of the local zonotopic estimates is ultimately bounded. This implies that the performance degradation of a local estimate has a limited impact on the performance of the DZFEs.

C. Related Works

It is widely recognized that the design of fusion criteria plays a central role in distributed fusion estimation for multi-sensor systems. Traditional fusion criteria focus on the fusion of Gaussian posterior estimates in a probabilistic framework. According to the availability of local estimation error correlations, they can be classified into two categories: C1) fusion criteria with known correlations; C2) fusion criteria with unknown correlations. For the first category, optimal fusion criteria can be designed by weighting the local estimates. In [26], an optimal fusion criterion in the maximum likelihood sense was designed for Gaussian posterior estimates. Subsequently, unified optimal fusion criteria for general linear data models, including raw measurement data and local estimates, were proposed in the best linear unbiased estimation sense and the weighted least squares sense [15]. Inspired by the work in [26], optimal fusion criteria weighted by scalars [27], diagonal matrices [28], and general matrices [29] were developed in the linear unbiased minimum variance sense. It was proven that the optimal fusion criterion weighted by general matrices [28] is equivalent to the maximum likelihood fusion criterion [26] for Gaussian posterior estimates, and also equivalent to the weighted least squares fusion criterion [15]. For the second category, the fusion criteria should yield consistent estimates regardless of the actual correlations. The well-known covariance intersection (CI) algorithm was proposed in [30] by using the convex combination of local estimates' covariances. On this basis, several variants of the CI algorithm have been proposed: fast CI algorithms [31], [32] to accelerate the optimization of weighting coefficients; split CI algorithms [33] to handle both known independent information and unknown correlations; sequential CI algorithms [34] for sequential fusion estimation; and inverse CI algorithms [35] for specific correlation structures involving unknown common information. Beyond fusion criteria for Gaussian posterior estimates, probability density function fusion has also received considerable attention in recent years. For example, CI technique was associated with log-linear combination of Gaussian functions in [36]–[39], while the fusion of general density functions was investigated in [40], [41].

On the other hand, set-membership estimation techniques are becoming increasingly popular especially in scenarios where noise and uncertainty are bounded but not probabilistically characterized. A variety of set-membership estimators have been developed based on the properties of different set representations. For example, intervals are used to design interval observers [42]–[46] due to their computational efficiency in numerical implementation. Ellipsoids also offer low-cost computations and are used for developing different ellipsoidal estimators [47]–[51]. General polytopes can describe set domains with any shape but encounter complexity issues when enumerating edges and vertices in high-dimensional spaces [52]. To reduce computational complexity, certain special polytopes, such as parallelotopes [53], [54] and zonotopes [55]–[60], were used for designing set-membership estimators. Actually, there is a trade-off between computational complexity and representation granularity for different sets, and a balanced set representation is helpful to estimator design and implement. Particularly, zonotopes can represent more sophisticated set domains than intervals, ellipsoids, and parallelotopes. Additionally, zonotopes can be fully characterized by a vector and a rectangular matrix, which simplifies zonotope operations to matrix calculations. Consequently, zonotopic estimation has gained increasing attention in recent research [22], [61]–[65]. This motivates the investigation of set-membership fusion estimation, especially zonotopic fusion estimation, to develop a new framework for multi-sensor systems. To date, research on fusion estimation under unknown but bounded noises is still in its early stages. A distributed fusion estimator under bounded noises with unknown lower and upper bounds was proposed in [66]. Meanwhile, the ellipsoidal fusion estimation problem was addressed in [67] using convex optimization,

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and zonotopic fusion estimators were developed by minimal parallelotope enclosure [24] and matrix-weighted fusion [23]. How to design fusion criteria to reduce the conservatism of fusion estimators remains an open question.

Notation: Throughout this paper, we use the following notations. The identity matrix is denoted by $\mathbf{I}$, and a column vector with all ones is represented as $\mathbf{1}$. Let us define $\mathbb{N}_+^L := \{1, 2, ..., L\}$, where L is a positive integer. Given sets Σ_1 and Σ_2, the notation $\Sigma_1 \setminus \Sigma_2$ denotes the set of elements in Σ_1 that are not in Σ_2. The notations $\mathrm{col}(a_1, a_2, \cdots, a_n)$ and $\mathrm{row}(a_1, a_2, \cdots, a_n)$ represent a column vector and a row vector, respectively, with elements $a_1, a_2, \cdots, a_n$. The trace, transpose, and inverse of a matrix A are represented by $\mathrm{Tr}(A)$, A^{T}, and A^{-1}, respectively. The matrix A is said to be 'spd' if it is symmetric ($A = A^{\mathrm{T}}$) and positive definite ($A \succ 0$, i.e., $\forall x \neq 0$, $x^{\mathrm{T}}Ax > 0$). Moreover, the infinite norm of a vector x is denoted by $\|x\|_\infty$, and the weighted Frobenius norm of a matrix $R \in \mathbb{R}^{n\times r}$ is defined as $\|R\|_W := \sqrt{\mathrm{Tr}(R^{\mathrm{T}}WR)}$, where $W \in \mathbb{R}^{n\times n}$ is a positive definite matrix. The notation $|\cdot|$ represents the element-by-element absolute value operator, and $\mathrm{diag}(x)$ is a diagonal matrix with diagonal elements from the vector x. Given a matrix $R \in \mathbb{R}^{n\times r}$, the submatrix consisting of columns a through b is denoted as $R_{:,a:b}$ (with $R_{:,a}$ representing the ath column), and $\vec{R}$ is the matrix obtained by sorting the columns of R in decreasing order of their weighted vector norm, such that $\left\|\vec{R}_{:,i}\right\|_W^2 \geq \left\|\vec{R}_{:,i+1}\right\|_W^2$.

II. Preliminaries

In this section, we will first introduce four representations of a zonotope. The general-representation of a zonotope $\mathcal{Z} \subset \mathbb{R}^n$ of order $\frac{r_0}{n}$ is given by:

$$\mathcal{Z} = \langle c, R\rangle := \{z \in \mathbb{R}^n : z = c + Ru,\ \|u\|_\infty \leq 1\} \tag{1}$$

where $c \in \mathbb{R}^n$ and $R \in \mathbb{R}^{n\times r_0}$ are the center and the generator matrix of $\mathcal{Z}$, respectively. Actually, the zonotope $\mathcal{Z}$ is an affine transformation of the unit hypercube $\mathcal{B}^{r_0} := \{u \in \mathbb{R}^{r_0} : \|u\|_\infty \leq 1\} \subset \mathbb{R}^{r_0}$. As a special kind of polytope, zonotopes also have $\mathcal{H}$-representations and $\mathcal{V}$-representations [52]. The $\mathcal{H}$-representation describes the zonotope $\mathcal{Z}$ as the intersection of closed half-spaces:

$$\mathcal{Z} = \{z \in \mathbb{R}^n : Hz \leq b\}. \tag{2}$$

Since zonotopes are centrally symmetric, we can express a pair of half-spaces corresponding to hyperplanes with the same normal vector as $\{z \in \mathbb{R}^n : h_i \leq b_{2i-1}\}$ and $\{z \in \mathbb{R}^n : -h_i \leq b_{2i}\}$. Then $H = \mathrm{col}\{h_1, -h_1, \cdots, h_i, -h_i, \cdots, h_\varepsilon, -h_\varepsilon\} \in \mathbb{R}^{2\varepsilon\times n}$ and $b = \mathrm{col}\{b_1, b_2, \cdots, b_{2i-1}, b_{2i}, \cdots, b_{2\varepsilon-1}, b_{2\varepsilon}\} \in \mathbb{R}^{2\varepsilon}$, where 2ε is the number of half-spaces.

The $\mathcal{V}$-representation expresses the zonotope $\mathcal{Z}$ as the convex hull of a finite set of vertices:

$$\begin{aligned}\mathcal{Z} &= \mathrm{conv}(V)\\ &= \left\{z \in \mathbb{R}^n : z = \sum\nolimits_{i=1}^{\kappa} \pi_i\nu_i, \pi_i \geq 0, \sum\nolimits_{i=1}^{\kappa} \pi_i = 1\right\}\end{aligned} \tag{3}$$

where $\nu_i \in \mathbb{R}^{n\times 1}$ is a vertex of the zonotope $\mathcal{Z}$, κ is the number of vertices, and $V = [\nu_1, ..., \nu_i, ..., \nu_\kappa] \in \mathbb{R}^{n\times\kappa}$. Moreover, the strip-representation of the zonotope $\mathcal{Z}$ can be obtained from (2) as the intersection of ε tight strips:

$$\mathcal{Z} = \bigcap\nolimits_{i=1}^{\varepsilon} \mathcal{S}(c_i, d_i) \tag{4}$$

where $\mathcal{S}(c_i, d_i) := \{z \in \mathbb{R}^n : |c_i z - d_i| \leq 1\}$ denotes a strip, a region enclosed by two parallel hyperplanes. Its parameters are calculated as $c_i = \frac{2h_i}{b_{2i-1}+b_{2i}}$ and $d_i = \frac{b_{2i-1}-b_{2i}}{b_{2i-1}+b_{2i}}$.

Then, we will introduce the following operations for zonotopes. The Minkowski sum $\oplus$ of two zonotopes and the linear image $\odot$ of a zonotope by a matrix L can be computed as follows [59]:

$$\langle c_1, R_1\rangle \oplus \langle c_2, R_2\rangle = \langle c_1 + c_2, [R_1 \quad R_2]\rangle \tag{5a}$$

$$L \odot \langle c, R\rangle = \langle Lc, LR\rangle. \tag{5b}$$

The order of the zonotope $\mathcal{Z}$ increases rapidly with consecutive Minkowski sum operations, leading to higher demands on storage and computational resources. This issue can be mitigated using a weighted reduction operation such that:

$$\langle c, R\rangle \subset \langle c, \downarrow_{r,W}(R)\rangle \tag{6}$$

where $\downarrow_{r,W}(\cdot) : \mathbb{R}^{n\times r_0} \to \mathbb{R}^{n\times r}$ is a weighted reduction operator [59] defined as

$$\downarrow_{r,W}(R) := \left[\vec{R}_{:,1:r-n},\ \mathrm{box}\left(\vec{R}_{:,r-n+1:r_0}\right)\right]. \tag{7}$$

Here, W is a given spd matrix, and r $(n \leq r \leq r_0)$ specifies the number of columns in the new generator matrix after the weighted reduction operation. This operator reorders the columns of R in decreasing order of their weighted vector norm as a new matrix $\vec{R}$, and then encloses the submatrix $\vec{R}_{:,r-n+1:r_0}$, containing the less important columns of R, into $\mathrm{box}(\vec{R}_{:,q-n+1:r})$. The aligned box of a matrix $R \in \mathbb{R}^{n\times r_0}$ is defined as

$$\mathrm{box}(R) := \begin{cases}\mathrm{diag}(|R| \cdot \mathbf{1}) & \text{if } r_0 > n\\ R & \text{otherwise.}\end{cases} \tag{8}$$

Remark 1. At least four order-reduction methods have been proposed to compute a lower-order zonotope that contains the original one [68]. Their core steps typically include reordering the generator matrix, partitioning the reordered matrix, overapproximating the split matrix, and finally overapproximating the zonotope. In [69], for example, the $r_0 - r + n$ generators with the smallest values of the score $\|R_{:,i}\|_1 - \|R_{:,i}\|_\infty$ are selected to be overapproximated by an aligned box. Although this reordering strategy can slightly reduce the approximation error [68], obtaining a quantitative error bound is significantly more challenging compared to the order reduction method used in this paper. The other two reduction methods approximate the split matrix using parallelotopes [68], [70], but they incur higher computational complexity.

Remark 2. In particular, the zonotopic estimation framework has advantages in reduced computational complexity and minimized wrapping effects caused by the propagation of overestimates [71]. The wrapping effect in the zonotope propagation can be effectively controlled as the operations of Minkowski sum and linear image are closed for zonotopes.

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The following lemma establishes the convexity of the function of weighted Frobenius norm square.

Lemma 1. Given a spd matrix $W \in \mathbb{R}^{n\times n}$, the function of weighted Frobenius norm square $f(\cdot) = \|\cdot\|_W^2$ with $\mathbf{dom}\ f = \mathbb{R}^{n\times m}$ is convex.

Proof. Let $X, Y \in \mathbb{R}^{n\times m}$ and $0 \le \theta \le 1$. The definition of the weighted Frobenius norm shows that

$$\begin{aligned}\|X\|_W^2 &= \mathrm{Tr}\left\{X^{\mathrm{T}}WX\right\} = \mathrm{Tr}\left\{XX^{\mathrm{T}}W\right\}\\ &= \mathrm{Tr}\left\{\sum_{i=1}^{m}\left(X_{:,i}X_{:,i}^{\mathrm{T}}W\right)\right\}\\ &= \sum_{i=1}^{m} X_{:,i}^{\mathrm{T}}WX_{:,i}\end{aligned} \tag{9}$$

and the convexity of $f(\cdot)$ follows from

$$\begin{aligned}&f(\theta X + (1-\theta)Y) - \theta f(X) - (1-\theta)f(Y)\\ &= -\theta(1-\theta)\sum_{i=1}^{m}\left((X_{:,i}-Y_{:,i})^{\mathrm{T}}W(X_{:,i}-Y_{:,i})\right) < 0.\end{aligned} \tag{10}$$

□

III. Problem Formulation

In this section, we formulate the problem of distributed zonotopic fusion estimation for multi-sensor systems. Consider a dynamic system monitored by L sensors (see Fig. 1 for a multi-sensor tracking system), where the state and measurement dynamics are described as follows:

$$x(k+1) = A(k)x(k) + B(k)w(k) \tag{11a}$$

$$y_i(k) = C_i(k)x(k) + D_i(k)v_i(k) \qquad (i \in \mathbb{N}_+^L). \tag{11b}$$

Here, $x(k) \in \mathbb{R}^n$ represents the system state, and $y_i(k) \in \mathbb{R}^{m_i}$ denotes the measurement from the ith sensor. The bounded matrices $A(k)$, $B(k)$, $C_i(k)$, and $D_i(k)$ have appropriate dimensions. The system noise $w(k)$, the measurement noise $v_i(k)$, and the initial state $x(0)$ are assumed to be wrapped within the following zonotopic sets:

$$\begin{cases} w(k) \in \mathbf{W} := \langle \mathbf{0}, \mathbf{I}_{n_w}\rangle\\ v_i(k) \in \mathbf{V}_i := \langle \mathbf{0}, \mathbf{I}_{n_{v_i}}\rangle\\ x(0) \in \langle c_0, R_0\rangle \subset \mathbb{R}^n.\end{cases} \tag{12}$$

At the sensor end, a local zonotopic estimate can be recursively designed—analogous to a Kalman filter—through three steps [24]: prediction, observation update, and reduction.

Prediction: Based on the state dynamics (11a), the prediction of $x(k)$ is as follows.

$$\begin{cases} \hat{\mathcal{X}}_i^p(k) := \langle \hat{x}_i^p(k), R_i^p(k)\rangle\\ \hat{x}_i^p(k) = A(k-1)\hat{x}_i(k-1)\\ R_i^p(k) = \left[A(k-1)R_i(k-1) \quad B(k-1)\right].\end{cases} \tag{13}$$

Observation Update: According to the measurement dynamics (11b), the observation update is designed using the current measurement $y_i(k)$.

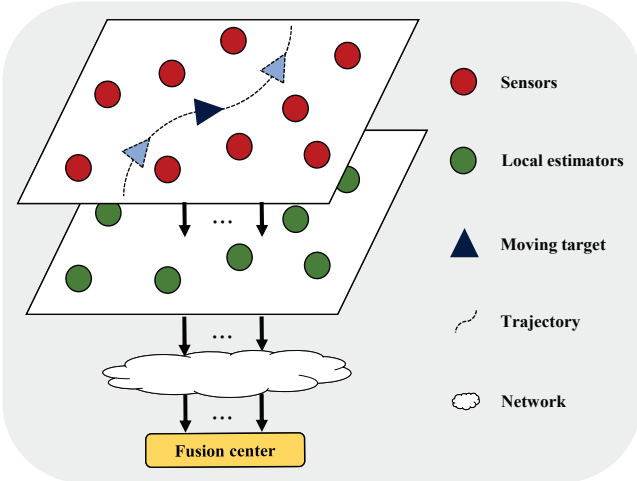

Fig. 1. A multi-sensor system for moving target tracking.

$$\begin{cases} \hat{\mathcal{X}}_i^o(k) := \langle \hat{x}_i^o(k), R_i^o(k)\rangle\\ \hat{x}_i^o(k) = \hat{x}_i^p(k) + K_i(k)\left[y_i(k) - C_i(k)\hat{x}_i^p(k)\right]\\ R_i^o(k) = \left[(\mathbf{I} - K_i(k)C_i(k))R_i^p(k) \quad -K_i(k)D_i(k)\right].\end{cases} \tag{14}$$

Reduction: To reduce computational complexity, the weighted reduction operator (7) is employed to keep the number of generators fixed at r $(n \le r)$:

$$\begin{cases} \hat{\mathcal{X}}_i(k) := \langle \hat{x}_i(k), R_i(k)\rangle\\ \hat{x}_i(k) = \hat{x}_i^o(k-1)\\ R_i(k) = \downarrow_{r,W}(R_i^o(k)).\end{cases} \tag{15}$$

The reduction step is crucial because the local zonotopic estimator operates on sensors with limited computational resources. Moreover, the optimal gain matrix $K_i^{\mathrm{opt}}(k)$, used throughout this paper, is determined as in [24], [59] by

$$\begin{aligned}K_i^{\mathrm{opt}}(k) =& R_i^p(k)\left[R_i^p(k)\right]^{\mathrm{T}} C_i^{\mathrm{T}}(k)\\ &\times\left\{C_i(k)R_i^p(k)\left[R_i^p(k)\right]^{\mathrm{T}} C_i^{\mathrm{T}}(k) + D_i(k)D_i^{\mathrm{T}}(k)\right\}^{-1}.\end{aligned} \tag{16}$$

It can also be proven that the state inclusion property $x(k) \in \hat{\mathcal{X}}_i(k)$ holds for all time instants if the initial estimate satisfies $x(0) \in \hat{\mathcal{X}}_i(0)$.

Afterwards, local zonotopic estimates $\hat{\mathcal{X}}_i(k)$ $(i \in \mathbb{N}_+^L)$ are transmitted from the sensor network to a FC via a communication network. In the FC, a zonotope fusion step combines locally computed estimates to yield a DZFE $\hat{\mathcal{X}}(k) := \langle \hat{x}(k), R(k)\rangle$ as follows:

$$\hat{\mathcal{X}}(k) = \mathcal{F}\left(\hat{\mathcal{X}}_1(k), ..., \hat{\mathcal{X}}_i(k), ..., \hat{\mathcal{X}}_L(k)\right). \tag{17}$$

Here, the function $\mathcal{F}(\cdot)$ represents the zonotope fusion criterion to be designed. The estimation performance of a zonotopic estimate $\hat{\mathcal{X}} := \langle c, R\rangle$ is evaluated based on the size of its wrapped region, which is fully characterized by its generator matrix. Therefore, the following performance index

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$J(\cdot)$ is defined and used in this paper to assess the estimation accuracy:

$$J\left(\hat{\mathcal{X}}\right) = J(R) := \|R\|_W^2. \tag{18}$$

Generally, the DZFE (17) is expected to outperform local zonotopic estimates in terms of estimation accuracy, i.e., $J\left(\hat{\mathcal{X}}(k)\right) \leq J\left(\hat{\mathcal{X}}_i(k)\right)$ $(i \in \mathbb{N}_+^L)$. Additionally, the DZFE (17) must satisfy the state inclusion property, i.e., $x(k) \in \hat{\mathcal{X}}(k)$, and its computational complexity should be manageable for real-time implementation. Consequently, the aims of this paper can be summarized as follows.

1) Design zonotope fusion criteria to ensure that the DZFEs provide better estimation performance than the local zonotopic estimates and maintain the state inclusion property.
2) Design sequential zonotope fusion criteria to reduce the computational complexity of DZFEs.
3) Derive stability conditions for the zonotopic fusion estimators to ensure that the estimation performance index of DZFEs is ultimately bounded.

Remark 3. The proposed DZFEs compute fusion estimates at a fusion center using local estimates collected from individual sensors. Compared with the consensus-based distributed estimation approach in [72], this distributed structure relies on the reliability of the fusion center but retains similar advantages in handling missing data and sensor faults. Local estimators operate independently at each sensor node and contribute their estimates to the fusion process. If some sensors lose communication capability, the remaining sensors can continue updating their local estimates and still contribute to the fusion step. Moreover, local fault detection and isolation mechanisms can be incorporated to identify and exclude faulty sensors without disrupting overall system functionality. For example, sensor faults can be detected by verifying whether the measurement falls within the predicted measurement region $\hat{\mathcal{Y}}_i^p(k) := \left(C_i(k) \odot \hat{\mathcal{X}}_i^p(k)\right) \oplus (D_i(k) \odot \mathbf{V}_i)$. This paper focuses on designing zonotope fusion criteria, but the results can be extended to centralized or consensus-based distributed structures.

Remark 4. Notice that the local zonotopic estimates satisfy the state inclusion property, meaning the system's real state is within the intersection of local zonotopic estimates. Although zonotopes are closed under the operations in (5), they are not closed under the intersection operation [22]. Therefore, the zonotope fusion criterion $\mathcal{F}(\cdot)$ is designed to find a zonotope enclosure of the intersection of local zonotopic estimates. Similar to the fusion criteria for Gaussian posterior estimates, a performance index $J(\cdot)$ is necessary for evaluating and optimizing the fusion results; however, the zonotope fusion criteria utilize the size of the zonotopes' wrapped region rather than the covariance matrix.

Remark 5. The constrained zonotope was introduced to combine the flexibility of convex polytopes with the efficiency and scalability of zonotopes [22], [62]. Although any zonotope can be expressed as a constrained zonotope, and the intersection of multiple constrained zonotopes remains a constrained zonotope, the resulting set is essentially a general polytope. Consequently, this representation does not directly address the zonotopic fusion problem, whose objective is to compute a zonotopic enclosure of the intersection of zonotopes. In other words, replacing zonotopes with constrained zonotopes does not solve the core challenge, as the outcome set does not preserve the zonotopic structure for efficient fusion.

IV. Main Results

This section presents the zonotope fusion criteria and sequential zonotope fusion criteria, followed by the derivation of stability conditions for the proposed DZFEs.

A. Zonotope fusion criterion

The technique for finding the intersection between zonotopes and strips is detailed in [57]. Based on this approach, we derive the following theorem as the first zonotope fusion criterion.

Theorem 1. Given local zonotopic estimates $\hat{\mathcal{X}}_i(k)$ $(i = 1, 2, ...L)$, a DZFE $\hat{\mathcal{X}}_{[1]}(k)$ can be determined by the following fusion criterion:

$$\begin{cases} \hat{x}_{[1]}(k) = \hat{x}_1(k) + \sum_{i=2}^{L} M_i(k)(\hat{x}_i(k) - \hat{x}_1(k)) \\ R_{[1]}(k) = \Bigg[\left(\mathbf{I} - \sum_{i=2}^{L} M_i(k)\right) R_1(k) \quad M_2(k)R_2(k) \\ \qquad\qquad\qquad\qquad \cdots \quad M_L(k)R_L(k)\Bigg] \end{cases} \tag{19}$$

where $M_i(k) \in \mathbb{R}^{n\times n}$ $(i = 2, ..., L)$ are designed parameter matrices. Moreover, the DZFE obtained from (19) satisfies $x(k) \in \mathcal{O}(k) \subseteq \hat{\mathcal{X}}_{[1]}(k)$, where $\mathcal{O}(k) := \bigcap_{i\in\mathbb{N}_+^L} \hat{\mathcal{X}}_i(k)$ is the intersection of the local zonotopic estimates.

Proof. Since $x(k) \in \hat{\mathcal{X}}_i(k)$ $(i \in \mathbb{N}_+^L)$, it follows that $x(k) \in \mathcal{O}(k)$. According to the general-representation of a zonotope, there exists $u_i(k) \in \mathcal{B}^r$ such that $x(k) = \hat{x}_i(k) + R_i(k)u_i(k)$ $(i = 1, 2, .., L)$. Equivalently, the following L equations must all be satisfied $\forall x(k) \in \mathcal{O}(k)$:

$$x(k) - \hat{x}_1(k) - R_1(k)u_1(k) = 0 \tag{20a}$$

$$\hat{x}_1(k) + R_1(k)u_1(k) - \hat{x}_i(k) - R_i(k)u_i(k) = 0 \tag{20b}$$

where $i = 2, 3, ..., L$. Note that (20) is a sufficient condition for the following equation:

$$x(k) = \hat{x}_{[1]}(k) + R_{[1]}(k)u(k) \tag{21}$$

where $u(k) := \mathrm{col}\{u_1(k), ..., u_L(k)\} \in \mathcal{B}^{rL}$. Again, using the general-representation of a zonotope, the equation (21) implies that $x(k) \in \hat{\mathcal{X}}_{[1]}(k)$. Hence, $\mathcal{O}(k) \subseteq \hat{\mathcal{X}}_{[1]}(k)$ is derived from (20) and (21). □

Remark 6. Although a zonotope can be represented as the intersection of strips, it is conservative to continuously perform the algorithm in [57] to obtain the intersection of zonotopes. In contrast, Theorem 1 fuses all zonotopes simultaneously to avoid potential conservatism. Actually, the parameter matrices $M_i(k)$ $(i = 2, ..., L)$ involve a total of $(L-1)n^2$ variables,

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This article has been accepted for publication in IEEE Transactions on Automatic Control. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TAC.2025.3647564

more than the $(L-1)n$ tunable variables used by the technique in [57].

Let us augment parameter matrices into $M(k) := \mathrm{row}\{M_2(k), ..., M_L(k)\}$. Then, the generator matrix in (19) can be rewritten as

$$R_{[1]}(k) = N_1(k) + M(k)N_2(k) \tag{22}$$

where the matrices $N_1(k)$ and $N_2(k)$ are defined by

$$N_1(k) = [R_1(k) \;\; \mathbf{0} \;\; \cdots \;\; \mathbf{0}] \tag{23a}$$

$$N_2(k) = \begin{bmatrix} -R_1(k) & R_2(k) & \mathbf{0} & \cdots & \mathbf{0} \\ -R_1(k) & \mathbf{0} & R_3(k) & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -R_1(k) & \mathbf{0} & \mathbf{0} & \cdots & R_L(k) \end{bmatrix}. \tag{23b}$$

To optimize the estimation performance of the designed DZFE, the following optimization problem with an analytical solution is formulated based on the performance index defined in (18).

Lemma 2. The optimal parameter matrices $M_i^{\mathrm{opt}}(k)$ $(i = 2..., L)$, which enable the designed DZFE (19) to achieve optimal estimation performance as defined in (18), can be determined by solving the following optimization problem:

$$\begin{aligned} M^{\mathrm{opt}}(k) &= \arg\min_{M(k)} J\left(R_{[1]}(k)\right) \\ &= \arg\min_{M(k)} \left\|R_{[1]}(k)\right\|_W^2. \end{aligned} \tag{24}$$

Moreover, the analytical solution to this optimization problem is given by

$$M^{\mathrm{opt}}(k) = -N_1(k)N_2^{\mathrm{T}}(k)\left(N_2(k)N_2^{\mathrm{T}}(k)\right)^{-1}. \tag{25}$$

Proof. By applying the definition of weighted Frobenius norm and (22), the derivative of the optimization objective (24) with respect to $M(k)$ is:

$$\begin{aligned} \frac{\partial \left\|R_{[1]}(k)\right\|_W^2}{\partial M(k)} = &\frac{\partial \mathrm{Tr}\left\{N_2^{\mathrm{T}}(k)M^{\mathrm{T}}(k)WM(k)N_2(k)\right\}}{\partial M(k)} \\ &+ \frac{\partial \mathrm{Tr}\left\{N_1^{\mathrm{T}}(k)WM(k)N_2(k)\right\}}{\partial M(k)} \\ &+ \frac{\partial \mathrm{Tr}\left\{N_2^{\mathrm{T}}(k)M^{\mathrm{T}}(k)WN_1(k)\right\}}{\partial M(k)}. \end{aligned} \tag{26}$$

Then, using the properties of trace derivatives, we obtain

$$\begin{aligned} \frac{\partial \left\|R_{[1]}(k)\right\|_W^2}{\partial M(k)} =& 2WM(k)N_2(k)N_2^{\mathrm{T}}(k) \\ &+ 2WN_1(k)N_2^{\mathrm{T}}(k). \end{aligned} \tag{27}$$

The convexity of $\left\|R_{[1]}(k)\right\|_W^2$ with respect to $M(k)$ follows from Lemma 1 and (22). Therefore, the minimum value of $\left\|R_{[1]}(k)\right\|_W^2$ is achieved when $\partial \left\|R_{[1]}(k)\right\|_W^2 / \partial M(k) = 0$. Since $R_i(k)$ has full row rank (due to the weighted reduction operation in (15), $N_2(k)$ also has full row rank, ensuring $N_2(k)N_2^{\mathrm{T}}(k)$ invertible. The analytical solution (25) follows from setting $\partial \left\|R_{[1]}(k)\right\|_W^2 / \partial M(k) = 0$. □

Remark 7. While the volume of a zonotopic estimate can also be used to evaluate the size of its wrapped region, computing this volume is computationally intensive. If volume were chosen as the estimation performance index, the optimization problem in (24) would lack an analytical solution and requires significantly more computing resources to solve. In contrast, the Frobenius norm is computationally efficient and allows for the derivation of an analytical solution to the proposed optimization problem.

Lemma 3. With the parameter matrices determined in Lemma 2, the DZFE $\hat{\mathcal{X}}_{[1]}^{\mathrm{opt}}(k) := \langle \hat{x}_{[1]}^{\mathrm{opt}}(k), R_{[1]}^{\mathrm{opt}}(k)\rangle$ provides better estimation performance than any original local zonotopic estimates, i.e.,

$$\left\|R_{[1]}^{\mathrm{opt}}(k)\right\|_W^2 \leq \|R_i(k)\|_W^2 \quad (i = 1, 2, ..., L) \tag{28}$$

where $R_{[1]}^{\mathrm{opt}}(k) := \min_{M(k)} \left\|R_{[1]}(k)\right\|_W^2$ can be obtained by substituting (25) into (22).

Proof. The proof of this Lemma is straightforward. If $M_i(k) = \mathbf{I}$ $(i \in \mathbb{N}_+^L \setminus \{1\})$ and $M_j(k) = \mathbf{0}$ $(j \in \mathbb{N}_+^L \setminus \{1, i\})$, then $\left\|R_{[1]}(k)\right\|_W^2 = \|R_i(k)\|_W^2$. If $M_i(k) = \mathbf{0}$ $(i \in \mathbb{N}_+^L \setminus \{1\})$, then $\left\|R_{[1]}(k)\right\|_W^2 = \|R_1(k)\|_W^2$. Notice that $R_{[1]}^{\mathrm{opt}}(k)$ with the parameter matrices obtained in (24) are optimal; thus (28) holds. □

Then, the computation procedures for the DZFE $\hat{\mathcal{X}}_{[1]}^{\mathrm{opt}}(k)$ are summarized in Algorithm 1.

Algorithm 1 Distributed zonotopic fusion estimation ($\hat{\mathcal{X}}_{[1]}^{\mathrm{opt}}(k)$).

Initialization
Given $\hat{\mathcal{X}}_i(0)$ $(i = 1, 2, \cdots, L)$ such that $x(0) \in \hat{\mathcal{X}}_i(0)$;
for $k = 1, 2, ...$ **do**
  Local Zonotopic Estimation
  Each sensor computes local zonotopic estimate $\hat{\mathcal{X}}_i(k)$ as in (13-15) by its local measurement $y_i(k)$;
  Zonotope Fusion
  The FC aggregates local zonotopic estimates $\hat{\mathcal{X}}_i(k)$ $(i = 1, 2, \cdots, L)$;
  The FC computes the analytical solution $M^{\mathrm{opt}}(k)$ using (25) for the optimization problem (24);
  The FC computes the DZFE $\hat{\mathcal{X}}_{[1]}^{\mathrm{opt}}(k)$ by (19);
end for

Remark 8. Different from the matrix-weighted distributed fusion estimators in [23], the proposed local estimates do not require knowing the initial state $x(0)$ and specifying the center $\hat{x}_i(k)$ at every instants according to the propagation of the state from $x(0)$. More importantly, the proposed DZFE $\hat{\mathcal{X}}_{[1]}(k)$ is based on finding a zonotope enclosure of the intersection of local zonotopic estimates. Thus, the state inclusion property can be strictly guaranteed for $\hat{\mathcal{X}}_{[1]}(k)$.

B. Sequential zonotope fusion criterion

The proposed zonotope fusion criterion in Theorem 1 uses a batch fusion strategy, which requires that all local zonotopic estimates be available at the FC before the fusion step can be

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implemented. Since the specific data arrival times at the FC vary among sensors during the period from instant $k-1$ to instant k, batch fusion strategies may induce long computation delays [25] and are thus unsuitable for real-time applications. Therefore, a sequential zonotope fusion criterion must be designed by combining the local zonotopic estimates one by one according to their arrival time order. The diagrams of batch and sequential zonotope fusion strategies are shown in Fig. 2.

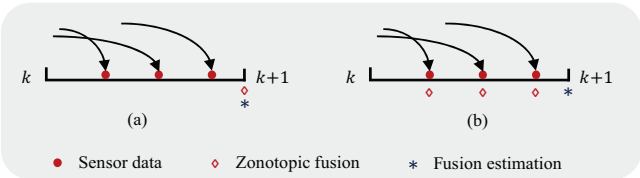

Fig. 2. The diagrams of two fusion strategies: (a) batch zonotope fusion; (b) sequential zonotope fusion.

Without loss of generality, we suppose that L local zonotopic estimates arrive in time order as $\hat{\mathcal{X}}_{\vec{1}}(k)$, ..., $\hat{\mathcal{X}}_{\vec{i}}(k)$, ..., $\hat{\mathcal{X}}_{\vec{L}}(k)$, where $\hat{\mathcal{X}}_{\vec{i}}(k) := \langle \hat{x}_{\vec{i}}(k), R_{\vec{i}}(k)\rangle$ is the ith arriving local zonotopic estimate, satisfying $\hat{\mathcal{X}}_{\vec{i}}(k) \in \left\{\hat{\mathcal{X}}_1(k), \hat{\mathcal{X}}_2(k), ..., \hat{\mathcal{X}}_L(k)\right\}$. The sequential zonotope fusion criterion comprises L fusion stages. The initial fusion result is $\hat{\mathcal{X}}_{f_1}(k) = \hat{\mathcal{X}}_{\vec{1}}(k)$, and subsequent fusion stages are represented as

$$\hat{\mathcal{X}}_{f_i}(k) = f\left(\hat{\mathcal{X}}_{f_{i-1}}(k), \hat{\mathcal{X}}_{\vec{i}}(k)\right) \quad (i = 2, 3, ..., L) \tag{29}$$

where $\hat{\mathcal{X}}_{f_i}(k) := \langle \hat{x}_{f_i}(k), R_{f_i}(k)\rangle$ $(i \in \mathbb{N}_+^L)$ is the ith fused result. The proposed sequential DZFE is the result of the final fusion stage $\hat{\mathcal{X}}_{[2]}(k) := \hat{\mathcal{X}}_{f_L}(k)$. The function $f(\cdot)$ represents a sequential zonotope fusion criterion, which is designed in the following theorem.

Theorem 2. Given the local zonotopic estimates arriving in time order $\hat{\mathcal{X}}_{\vec{i}}(k)$ $(i = 2, ...L)$, the ith sequential fusion stage defined in (29) yields $\hat{\mathcal{X}}_{f_i}(k)$ as follows:

$$\begin{cases} \hat{x}_{f_i}(k) = \hat{x}_{f_{i-1}}(k) + M_i(k)\left(\hat{x}_{\vec{i}}(k) - \hat{x}_{f_{i-1}}(k)\right) \\ R_{f_i}(k) = \left[\,(I - M_i(k))R_{f_{i-1}}(k) \quad M_i(k)R_{\vec{i}}(k)\right] \end{cases} \tag{30}$$

where $M_i(k) \in \mathbb{R}^{n\times n}$ is a designed parameter matrix. Moreover, the sequential DZFE $\hat{\mathcal{X}}_{[2]}(k) = \hat{\mathcal{X}}_{f_L}(k)$ satisfies the state inclusion property, i.e., $x(k) \in \mathcal{O}(k) \subseteq \hat{\mathcal{X}}_{[2]}(k)$.

Proof. We first prove that $\hat{\mathcal{X}}_{f_{i-1}}(k) \bigcap \hat{\mathcal{X}}_{\vec{i}}(k) \subseteq \hat{\mathcal{X}}_{f_i}(k)$ for the ith fusion stage. By the general-representation of a zonotope, if $x(k) \in \hat{\mathcal{X}}_{f_{i-1}}(k) \bigcap \hat{\mathcal{X}}_{\vec{i}}(k)$, then there exist $u_{f_{i-1}}(k) \in \mathcal{B}^{r(i-1)}$ and $u_{\vec{i}}(k) \in \mathcal{B}^r$ such that

$$x(k) - \hat{x}_{f_{i-1}}(k) - R_{f_{i-1}}(k)u_{f_{i-1}}(k) = 0 \tag{31a}$$

$$\hat{x}_{f_{i-1}}(k) + R_{f_{i-1}}(k)u_{f_{i-1}}(k) - \hat{x}_{\vec{i}}(k) - R_{\vec{i}}(k)u_{\vec{i}}(k) = 0. \tag{31b}$$

The equations in (31) are a sufficient condition for the following equation:

$$x(k) = \hat{x}_{f_i}(k) + R_{f_i}(k)u_{f_i}(k) \tag{32}$$

where $u_{f_i}(k) := \mathrm{col}\{u_{f_{i-1}}(k), u_{\vec{i}}(k)\} \in \mathcal{B}^{ri}$. It means that $x(k) \in \hat{\mathcal{X}}_{f_i}(k)$ by the general-representation of a zonotope, and $\hat{\mathcal{X}}_{f_{i-1}}(k) \bigcap \hat{\mathcal{X}}_{\vec{i}}(k) \subseteq \hat{\mathcal{X}}_{f_i}(k)$ can be inferred from (31) to (32). We now proceed with the proof by induction. It is known from the initial fusion stage that $x(k) \in \hat{\mathcal{X}}_{\vec{1}}(k) = \hat{\mathcal{X}}_{f_1}(k)$. For the ith fusion stage in (30), assume that $x(k) \in \bigcap_{j=1}^{i-1} \hat{\mathcal{X}}_{\vec{j}}(k) \subseteq \hat{\mathcal{X}}_{f_{i-1}}(k)$, then $x(k) \in \bigcap_{j=1}^{i} \hat{\mathcal{X}}_{\vec{j}}(k) \subseteq \hat{\mathcal{X}}_{f_{i-1}}(k) \bigcap \hat{\mathcal{X}}_{\vec{i}}(k) \subseteq \hat{\mathcal{X}}_{f_i}(k)$ holds by previous derivation. By induction, we conclude that $x(k) \in \mathcal{O}(k) = \bigcap_{i=1}^{L} \hat{\mathcal{X}}_{\vec{i}}(k) \subseteq \hat{\mathcal{X}}_{f_L}(k) = \hat{\mathcal{X}}_{[2]}(k)$. □

Lemma 4. The optimal parameter matrices $M_i^{\mathrm{opt}}(k)$ $(i = 1, 2..., L)$, which enable the fusion results in (30) to achieve the optimal estimation performance (18) at every stages, can be determined by the following optimization problem:

$$\begin{aligned} M_i^{\mathrm{opt}}(k) &= \arg\min_{M_i(k)} J\left(R_{f_i}(k)\right) \\ &= \arg\min_{M_i(k)} \|R_{f_i}(k)\|_W^2 . \end{aligned} \tag{33}$$

Moreover, the analytical solution of (33) is given by

$$M_i^{\mathrm{opt}}(k) = R_{f_{i-1}}(k)R_{f_{i-1}}^{\mathrm{T}}(k)\left(R_{f_{i-1}}(k)R_{f_{i-1}}^{\mathrm{T}}(k) + R_{\vec{i}}(k)R_{\vec{i}}^{\mathrm{T}}(k)\right)^{-1}. \tag{34}$$

Proof. Similar to the proof of Lemma 2, the derivative of the optimization objective (33) with respect to $M_i(k)$ is:

$$\begin{aligned} \frac{\partial \|R_{f_i}(k)\|_W^2}{\partial M_i(k)} &= \frac{\partial \left\|(I - M_i(k))\, R_{f_{i-1}}(k)\right\|_W^2}{\partial M_i(k)} \\ &\quad + \frac{\partial \left\|M_i(k)R_{\vec{i}}(k)\right\|_W^2}{\partial M_i(k)}. \end{aligned} \tag{35}$$

By the definition of the weighted Frobenius norm and the properties of trace derivatives, we obtain

$$\begin{aligned} \frac{\partial \|R_{f_i}(k)\|_W^2}{\partial M_i(k)} &= 2WM_i(k)R_{f_{i-1}}(k)R_{f_{i-1}}^{\mathrm{T}}(k) \\ &\quad - 2WR_{f_{i-1}}(k)R_{f_{i-1}}^{\mathrm{T}}(k) + 2WM_i(k)R_{\vec{i}}(k)R_{\vec{i}}^{\mathrm{T}}(k). \end{aligned} \tag{36}$$

It is known from Lemma 1 and (30) that $\|R_{f_i}(k)\|_W^2$ is convex with respect to $M_i(k)$. Therefore, the optimal estimation performance is achieved when $\partial \|R_{f_i}(k)\|_W^2 / \partial M_i(k) = 0$, and the analytical solution in (34) follows from (36). □

Lemma 5. With the parameter matrices determined in Theorem 2, the sequential DZFE $\hat{\mathcal{X}}_{[2]}^{\mathrm{opt}}(k) := \langle \hat{x}_{[2]}^{\mathrm{opt}}(k), R_{[2]}^{\mathrm{opt}}(k)\rangle$ in (29-30) with optimal parameters in (33) achieves better estimation performance than any original local zonotopic estimates, i.e.,

$$\left\|R_{[2]}^{\mathrm{opt}}(k)\right\|_W^2 \le \|R_i(k)\|_W^2 \quad (i = 1, 2, ..., L). \tag{37}$$

Proof. The proof is similar to the derivation of Lemma 3, and is therefore omitted. □

The computation procedures for the sequential DZFE $\hat{\mathcal{X}}_{[2]}^{\mathrm{opt}}(k)$ are summarized in Algorithm 2.

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Algorithm 2 Sequential zonotopic fusion estimation.

Initialization
Given $\hat{\mathcal{X}}_i(0)$ $(i = 1, 2, \cdots, L)$ such that $x(0) \in \hat{\mathcal{X}}_i(0)$;
for $k = 1, 2, \ldots$ **do**
  Local Zonotopic Estimation
  Each sensor computes local zonotopic estimate $\hat{\mathcal{X}}_i(k)$ as in (13-15) by its local measurement $y_i(k)$;
  Zonotope Fusion
  The FC aggregates local zonotopic estimates and orders them by arrival time as $\hat{\mathcal{X}}_{\tilde{i}}(k)$ $(i = 1, 2, \cdots, L)$;
  The FC initializes the fusion stage $\hat{\mathcal{X}}_{f_1}(k) = \hat{\mathcal{X}}_{\tilde{1}}(k)$;
  for $i = 2, \ldots, L$ **do**
    The FC computes the analytical solution (34) of the optimization problem (33) to obtain the optimal parameter matrices $M_i^{\text{opt}}(k)$;
    The FC computes the ith fusion result $\hat{\mathcal{X}}_{f_i}(k)$ by (30);
  end for
  The FC obtains the sequential DZFE $\hat{\mathcal{X}}_{[2]}^{\text{opt}}(k) = \hat{\mathcal{X}}_{f_L}(k)$.
end for

Remark 9. The computational complexity of (25) arises from a high-dimensional matrix inverse operation, which has a cost of $O\left((L-1)^{2.38}n^{2.38}\right)$. In contrast, the sequential fusion simplifies the computation into $L-1$ low-dimensional matrix inverse operations in (34), each with a cost of $O\left(n^{2.38}\right)$. As a result, the overall computational complexity is significantly reduced in the sequential DZFE. Moreover, the sequential fusion mechanism splits $L-1$ sequential fusion stages across the entire time horizon, thereby avoiding long computation delays.

Remark 10. Compared with the sequential CI Kalman fusion in [25], [34], the proposed sequential zonotope fusion is tailored for local zonotopic estimates. The fusion principle is to find a zonotope enclosure for two zonotopic estimates at each fusion stage rather than weighting Gaussian posterior estimates, while the optimization objective is the size of the fused generator matrix rather than the covariance matrix. A similar benefit is that the introduction of sequential fusion mechanism contributes to the reduction of the complexity and computational burden in fusion algorithms.

C. Improvement Strategy

Although the DZFE and the sequential DZFE with optimal parameters can provide better estimation performance than original local zonotopic estimates, the fusion result is still conservative. To further enhance the performance of the designed DZFEs, we propose an improvement strategy inspired by our previous work [24]. The improvement strategy is illustrated in Fig. 3, where the intersection and its zonotope enclosure are from the first example in Section V. Define $\hat{\mathcal{X}}^{\text{opt}}(k) := \langle \hat{x}^{\text{opt}}(k), R^{\text{opt}}(k)\rangle \in \left\{\hat{\mathcal{X}}_{[1]}^{\text{opt}}(k), \hat{\mathcal{X}}_{[2]}^{\text{opt}}(k)\right\}$, the $\mathcal{H}$-representation of $\hat{\mathcal{X}}^{\text{opt}}(k)$ can be written as

$$\hat{\mathcal{X}}^{\text{opt}}(k) = \{x(k) \in \mathbb{R}^n : H(k)x(k) \leq b(k)\} \tag{38}$$

where $H(k) \in \mathbb{R}^{2\varepsilon \times n}$ and $b(k) \in \mathbb{R}^{2\varepsilon}$ are denoted by

$$\begin{cases} H(k) = \text{col}\{h_1(k), -h_1(k), \cdots, h_\varepsilon(k), -h_\varepsilon(k)\} \\ b(k) = \text{col}\{b_1(k), \cdots, b_{2\varepsilon}(k)\}. \end{cases} \tag{39}$$

Obviously, the intersection of local zonotopic estimates $\mathcal{O}(k)$ is a polytope. We denote its $\mathcal{V}$-representation as:

$$\mathcal{O}(k) = \text{conv}(V(k)) \tag{40}$$

where $V(k) = \begin{bmatrix}\nu_1(k) & \cdots & \nu_\eta(k)\end{bmatrix}$. A pair of parallel hyperplanes (see Fig. 3 (b)) bounding $\hat{\mathcal{X}}^{\text{opt}}(k)$, i.e., $\{x(k) : h_i(k)x(k) = b_{2i-1}(k)\}$ and $\{x(k) : h_i(k)x(k) = -b_{2i}(k)\}$, may not constitute a pair of tight parallel hyperplanes for the intersection $\mathcal{O}(k)$. The proposed approach reduces the distance between these hyperplanes to make them a pair of tight parallel hyperplanes bounding $\mathcal{O}(k)$.

Given the center $\hat{x}^{\text{opt}}(k)$, the goal is to find strips with the normal vector $h_i(k)$ $(i \in \mathbb{N}_+^\varepsilon)$ that exactly enclose $\mathcal{O}(k)$. The first step is to identify a vertex of $\mathcal{O}(k)$ that is farthest from the center along the direction $h_i(k)$. This vertex is found by solving the following problem:

$$\mathbf{s}_i(k) = \arg\max_{j \in \mathbb{N}_+^\eta} \left|h_i(k)\left(\hat{x}^{\text{opt}}(k) - \nu_j(k)\right)\right|. \tag{41}$$

Then, the two parallel hyperplanes (Fig. 3 (c)) can be represented by

$$\mathcal{P}_{i,1}(k) := \left\{z(k) \in \mathbb{R}^n : h_i(k)z(k) \leq h_i(k)\nu_{\mathbf{s}_i(k)}(k)\right\} \tag{42a}$$

$$\mathcal{P}_{i,2}(k) := \Big\{z(k) \in \mathbb{R}^n : h_i(k)z(k) \leq 2h_i(k)\hat{x}^{\text{opt}}(k) - h_i(k)\nu_{\mathbf{s}_i(k)}(k)\Big\}. \tag{42b}$$

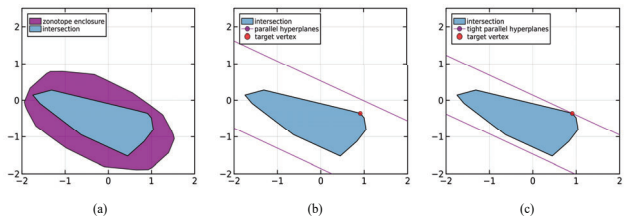

Fig. 3. (a) The intersection region and its zonotope enclosure; (b) the parallel hyperplanes and the target vertex.; (c) the tight parallel hyperplanes.

The parallel hyperplanes in (42) define a strip $\mathcal{S}(c_i(k), d_i(k))$ $(i \in \mathbb{N}_+^\varepsilon)$. The following theorem then constructs the improved zonotope fusion criterion.

Theorem 3. Given the DZFE $\hat{\mathcal{X}}^{\text{opt}}(k)$, an improved DZFE $\hat{\mathcal{X}}^{\text{imp}}(k) := \langle \hat{x}^{\text{imp}}(k), R^{\text{imp}}(k)\rangle$ can be determined by the following fusion criterion:

$$\hat{\mathcal{X}}^{\text{imp}}(k) = \bigcap_{i=1}^{\mathbb{N}_+^\varepsilon} \mathcal{S}(c_i(k), d_i(k)). \tag{43}$$

Moreover, the improved DZFE obtained by (43) satisfies $x(k) \in \mathcal{O}(k) \subseteq \hat{\mathcal{X}}^{\text{imp}}(k) \subseteq \hat{\mathcal{X}}^{\text{opt}}(k)$.

Proof. By the construction of strips $\mathcal{S}(c_i(k), d_i(k))$ $(i = 1, \ldots, \varepsilon)$ in (41) and (42), it is guaranteed that

$$\mathcal{O}(k) \subseteq \bigcap_{i=1}^{\mathbb{N}_+^\varepsilon} \mathcal{S}(c_i(k), d_i(k)). \tag{44}$$

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Therefore, it is inferred that $x(k) \in \mathcal{O}(k) \subseteq \hat{\mathcal{X}}^{\text{imp}}(k)$. Since the parallel hyperplanes in (42) are tight parallel hyperplanes for the intersection $\mathcal{O}(k)$, it follows that $\hat{\mathcal{X}}^{\text{imp}}(k) \subseteq \hat{\mathcal{X}}^{\text{opt}}(k)$. □

Lemma 6. The improved DZFE $\hat{\mathcal{X}}^{\text{imp}}(k)$ in Theorem 3 achieves better estimation performance than any original local zonotopic estimates, i.e.,

$$\left\|R^{\text{imp}}(k)\right\|_W^2 \leq \|R_i(k)\|_W^2 \quad (i = 1, 2, ..., L). \tag{45}$$

Proof. The proof of this lemma follows by the fact that $\hat{\mathcal{X}}^{\text{imp}}(k) \subseteq \hat{\mathcal{X}}^{\text{opt}}(k)$ and that the generator $R^{\text{opt}}(k)$ is reduced to $R^{\text{imp}}(k)$ after slicing the zonotope along the normal vectors of the tight hyperplanes. The result then follows from Lemmas 3 and 5. □

Remark 11. The improved DZFE satisfies the state inclusion property because $\mathcal{O}(k) \subseteq \hat{\mathcal{X}}^{\text{imp}}(k)$, meaning $\hat{\mathcal{X}}^{\text{imp}}(k)$ inherits the performance superiority and the state inclusion property from $\hat{\mathcal{X}}^{\text{opt}}(k)$. Compared with the fusion criterion in our previous work [24] using a parallelotope enclosure, the proposed fusion criterion in Theorem 3 offers additional performance superiority, making it more reliable and useful for zonotope fusion. Moreover, the fusion criterion in Theorem 3 is less conservative because the general zonotopes can describe more accurate sets than parallelotopes.

Remark 12. Given the general-representation of a zonotope $\langle c, R\rangle$, its $\mathcal{V}$-representation (vertices) can be enumerated by $\nu = c + \sum_{i=1}^{r} \alpha_i R_{:,i}$, $\alpha_i \in \{-1, +1\}$, and its $\mathcal{H}$-representation is obtained from these vertices using convex hull algorithms. Conversely, given the $\mathcal{H}$-representation of a polytope, its $\mathcal{V}$-representation can be computed by vertex enumeration algorithms. Common convex hull algorithms and vertex enumeration algorithms have been integrated into toolboxes such as 'cdd' [73].

D. Stability Analysis

To analyze the stability conditions for the proposed DZFEs, we first introduce relevant lemmas. Recall from (15) that the reduced generator matrix is defined as $R_i(k) = \downarrow_{r,W} (R_i^o(k))$. The following lemma from [59] shows the performance relationship before and after the weighted reduction operation.

Lemma 7. Given a spd matrix W_i with all its eigenvalues in $[\underline{\lambda}_i, \overline{\lambda}_i] \subset \mathbb{R}, \underline{\lambda}_i > 0$, the performance relationship between $R_i^o(k) \in \mathbb{R}^{n\times\bar{r}}$ and $R_i(k) \in \mathbb{R}^{n\times r}$ satisfies

$$\|R_i(k)\|_{W_i}^2 \leq \left(1 + \frac{\mu_i}{d+r}\right) \|R_i^o(k)\|_{W_i}^2 \tag{46}$$

where $d = \bar{r} - r$ and $\mu_i = ((\overline{\lambda}_i/\underline{\lambda}_i)(d+n) - 1)(d+n)$.

The following lemma from [59] provides conditions to keep a linear time-varying system robustly γ-stable.

Lemma 8. The time-varying discrete-time linear system $x(k) = A(k-1)x(k-1)$ is robustly γ-stable (i.e., stable with decay rate γ) if there exists a bounded sequence of spd matrices $W(k)$ and $\gamma \in [0, 1]$ such that, $\forall k \geq 0$,

$$\begin{bmatrix} \gamma W(k-1) & A^{\text{T}}(k-1)W(k) \\ \star & W(k) \end{bmatrix} \succ 0. \tag{47}$$

Assumption 1. A bounded sequence of local estimator gain matrices $K_i^*(k)$ and a bounded sequence of spd matrices $W_i^*(k)$ are assumed to be designed such that $x_i(k) = (\mathbf{I} - K_i^*(k)C_i(k))A(k-1)x_i(k-1)$ is robustly γ_i-stable, where $\gamma_i < 1$ and $W_i^*(k)$ $(\forall k > 0)$ have all its eigenvalues within a constant real interval $[\underline{\lambda}_i, \overline{\lambda}_i], \underline{\lambda}_i > 0$.

The condition in Assumption 1 is not strict. When $\gamma_i = 1$, the existing condition on $K_i^*(k)$ is much weaker than the uniformly observable and controllable conditions required for the standard Kalman filter. Given γ_i, the bounded sequences of matrices can be obtained by solving a linear matrix inequality problem, as clarified in [24].

The following lemma leverages the performance superiority of the proposed DZFEs.

Lemma 9. The proposed DZFEs in Theorems 1-2 are ultimately bounded if at least one of the local zonotopic estimates is ultimately bounded, i.e.,

$$\begin{aligned} &\lim_{k\to\infty} \|R_i(k)\|_{W_i^*(k)}^2 < p_i \ (\exists i \in \mathbb{N}_+^L) \\ &\qquad\qquad \Longrightarrow \lim_{k\to\infty} \|R(k)\|_{W_i^*(k)}^2 < p_i \end{aligned} \tag{48}$$

where p_i is a positive finite number.

Proof. By the performance superiority shown in Lemmas 3, 5, and 6, we have $\|R(k)\|_W^2 \leq \|R_i(k)\|_W^2$ $(i \in \mathbb{N}_+^L)$. The result then follows directly from this inequality. □

Theorem 4. The proposed DZFEs in Theorems 1-2 are ultimately bounded if there exists $i \in \mathbb{N}_+^L$ such that $\gamma_i \left(1 + \frac{\mu_i}{d+r}\right) < 1$ holds.

Proof. Similar to the analysis in [24], we can derive from Assumption 1 that if $\gamma_i \left(1 + \frac{\mu_i}{d+r}\right) < 1$ holds, then

$$\lim_{k\to\infty} \|R_i(k)\|_{W_i^*(k)}^2 < p_i. \tag{49}$$

By Lemma 9, this theorem is then proven. □

V. Numerical Examples

This section first demonstrates the advantages of the proposed zonotope fusion criteria by fusing two-dimensional zonotopes, and then evaluates the effectiveness of the proposed DZFEs using a target tracking example.

A. Fusion of Two-dimensional Zonotopes

In this example, three 2-dimensional zonotopes, each with four generators, are fused using different zonotope fusion criteria. The original zonotopes and their intersection are plotted in Fig. 4 (a). The minimum zonotope enclosure defined in (19) under the performance index (18) is plotted in Fig. 4 (b), computed via the analytical solution in Lemma 2.

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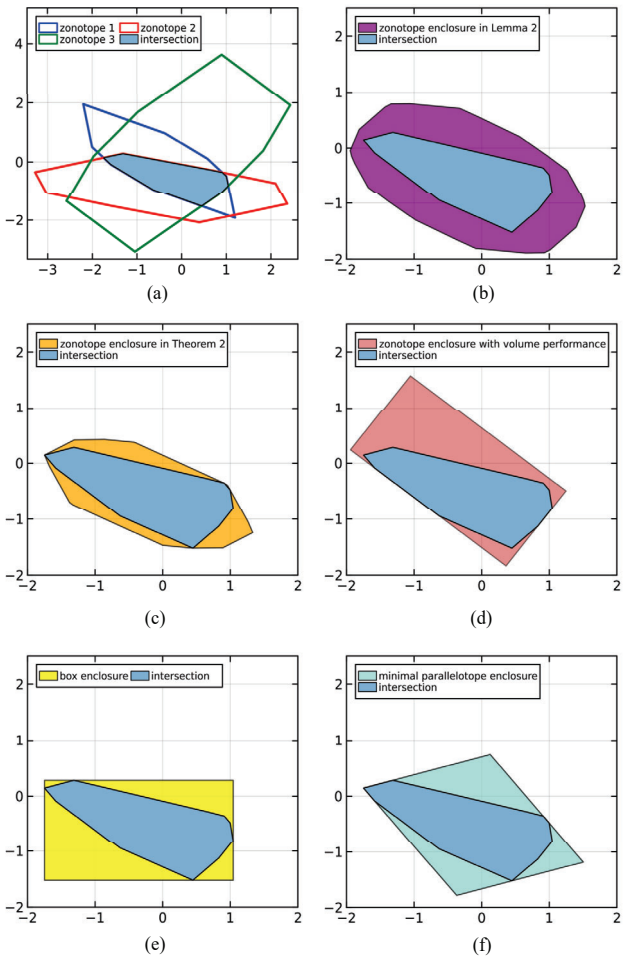

Fig. 4. (a) Three zonotopes and their intersection. (b) The minimal zonotope enclosure by Lemma 2. (c) The improved zonotope enclosure by Theorem 2. (d) The minimal zonotope enclosure under volume performance. (e)The box enclosure. (f) The minimum parallelotope enclosure.

To reduce conservatism, the improved zonotope enclosure is computed by Theorem 3 and plotted in Fig. 4 (c). Fig. 4 (b-c) clearly shows that the improvement strategy from Theorem 3 significantly reduces the size of the zonotope enclosure. Additionally, the minimum zonotope enclosure defined in (19) under the volume performance index is obtained by solving a nonlinear optimization problem and is presented in Fig. 4 (d). Although solving the nonlinear optimization problem requires significant computational effort, its solution is not guaranteed to be globally optimal. Therefore, the minimum volume zonotope enclosure does not offer any clear performance advantage over the improved zonotope fusion, as illustrated in Fig. 4 (d). Furthermore, the box enclosure and the minimum parallelotope enclosure from [24] are plotted in Fig. 4 (e-f). As discussed in Remark 11, these two enclosures do not provide performance superiority over the original zonotopes. To evaluate computational efficiency, 2-dimensional zonotopes with different numbers of generators are randomly generated. The computing time for the various zonotope fusion criteria are plotted in Fig. 5. As shown in Fig. 5 (a), the minimum zonotope enclosure under the volume performance index requires more time to solve the optimization problem, and its computing time increases exponentially with the number of generators, whereas the other two fusion criteria exhibit a much slower increase. To illustrate the advantages of sequential zonotope fusion, the total computing time for the sequential fusion stages is compared to the batch zonotope fusion in Lemma 2. The results, shown in Fig. 5 (b), indicate that sequential zonotope fusion requires less computational time, as it involves only low-dimensional matrix computation.

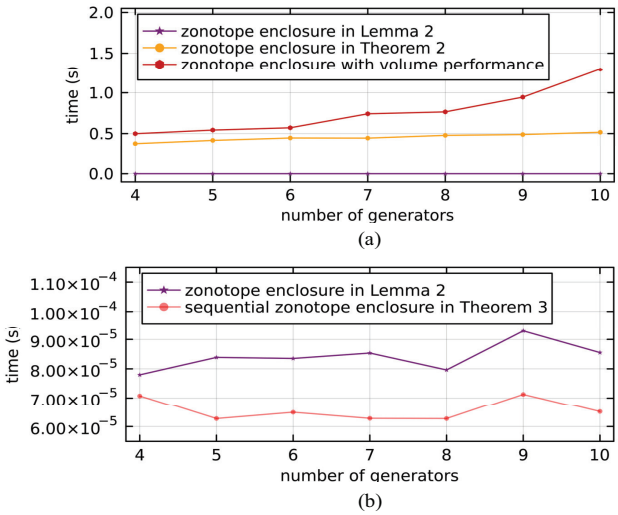

Fig. 5. The computing time of zonotope fusion criteria.

B. Target Tracking with Multi-sensors

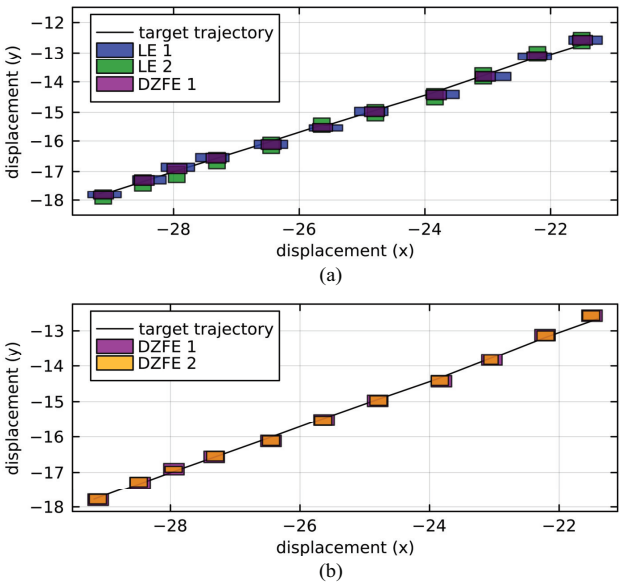

Fig. 6. The trajectory of the moving target and the projections of zonotopic estimates in (d_x, d_y): (a) two local zonotopic estimates and the DZFE with optimal parameters; (b) the DZFE with optimal parameters and the improved DZFE.

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TABLE I
THE ESTIMATION PERFORMANCE

Performance index	LE 1	LE 2	DZFE 1	DZFE 2	DZFE 3	DZFE 4	DZFE 5
$\|\cdot\|_W^2$	0.695	0.695	**0.413**	0.467	0.471	0.467	0.560
$\mathrm{Vol}(\cdot)$	0.0934	0.0934	0.0527	**0.0227**	0.0259	0.0228	0.0482

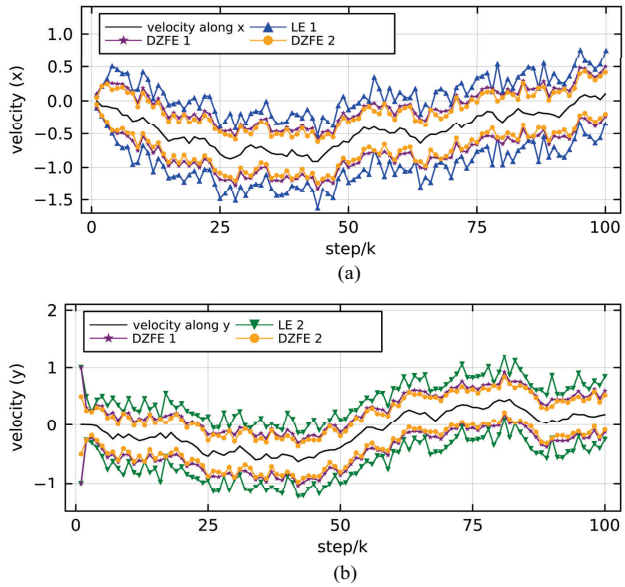

Fig. 7. The velocity of the moving target and the projections of the local zonotopic estimates, the DZFE with optimal parameters, and the improved DZFE: (a) along $\mathbf{x}$ direction; (b) along $\mathbf{y}$ direction.

In the second example, we consider a target tracking system with two sensors. The dynamics of a non-maneuvering target is described as follows [74]:

$$x(k+1)=\begin{bmatrix}1 & T & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & T\\ 0 & 0 & 0 & 1\end{bmatrix}x(k)+\begin{bmatrix}T^2/2 & 0\\ T & 0\\ 0 & T^2/2\\ 0 & T\end{bmatrix}w(k) \quad (50)$$

where $x := [d_{\mathbf{x}}\ v_{\mathbf{x}}\ d_{\mathbf{y}}\ v_{\mathbf{y}}]^{\mathrm{T}}$ represents the target's state, and $w(k)$ denotes the system's process noise. The parameter T is the sampling period, while $(d_{\mathbf{x}}, d_{\mathbf{y}})$ and $(v_{\mathbf{x}}, v_{\mathbf{y}})$ are the displacement and velocity components along the $\mathbf{x}$ and $\mathbf{y}$ axes, respectively. The target's displacement components are observed by two sensors with differing precision, and the measurement model is represented as follows:

$$\begin{cases} y_1(k) = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix}x(k) + \begin{bmatrix}2 & 0\\ 0 & 1\end{bmatrix}v_1(k) \\ y_2(k) = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix}x(k) + \begin{bmatrix}1 & 0\\ 0 & 2\end{bmatrix}v_2(k). \end{cases} \quad (51)$$

Here, $y_i(k)$ $(i = 1, 2)$ and $v_i(k)$ $(i = 1, 2)$ denote the measurements and the measurement noises, respectively.

In the simulation, the sampling period is set to $T = 1$. Both process noise and measurement noise are bounded within the zonotope $\langle \mathbf{0}, \mathbf{I} \rangle$. Two local zonotopic estimators are deployed to estimate the target's state. These local estimates are then transmitted to a FC via a communication network for zonotope fusion. Different zonotopic fusion estimates are computed based on different zonotope fusion criteria. DZFE 1 is the result of the analytical solution provided in Lemma 2, which is optimal under the performance index (18). DZFE 2 is computed from DZFE 1 using the improvement strategy in Theorem 3. To show the effectiveness of the proposed fusion estimators, the computed zonotopes are projected onto the displacement coordinates $(d_{\mathbf{x}}, d_{\mathbf{y}})$. Fig. 6 depicts the target trajectory and the projections of zonotopic estimates during time steps 40 to 50. The projection of DZFE 1 in $(d_{\mathbf{x}}, d_{\mathbf{y}})$ is significantly smaller than that of the local zonotopic estimates, and the projection of DZFE 2 is fully enclosed within that of DZFE 1. Both fusion estimators outperform the local estimators in estimation performance, with DZFE 2 exhibiting less conservatism than DZFE 1. Moreover, the computed zonotopes are projected onto the $v_{\mathbf{x}}$ and $v_{\mathbf{y}}$ coordinates, as shown in Fig. 7. It can be observed that the projected interval of DZFE 1 along the $\mathbf{x}$ direction is within that of the first local zonotopic estimate, while its projected interval along the $\mathbf{y}$ direction is within that of the second local zonotopic estimate. This aligns with the sensor configuration where the first sensor has better precision in the $\mathbf{y}$ direction, and the second sensor in the $\mathbf{x}$ direction. In line with the improvement strategy, the projected intervals of DZFE 2 fall within those of DZFE 1. In addition, DZFEs 3-5 are computed using the optimization problem in (24) under the volume performance index, the box enclosure, and the minimal parallelotope enclosure, respectively. The weighted Frobenius norms and volumes of these DZFEs are calculated, with the average values over time shown in Table I, where LE 1 and LE 2 represent the two local zonotopic estimates. The results indicate that all fusion estimates outperform local estimates, achieving superiority in both weighted Frobenius norm and volume. Particularly, DZFE 1 achieves the minimal weighted Frobenius norm, while DZFE 2 achieves the minimal volume, thereby illustrating the effectiveness of the improvement strategy. Notably, the volume performance of DZFE 2 is nearly identical to that of DZFE 4. This can be attributed to the particular structure of the measurement matrices and the presence of two dominant generators in the local zonotopic estimates.

VI. CONCLUSION

In this paper, we investigated the problem of distributed zonotopic fusion estimation. Three zonotope fusion criteria were designed to integrate local zonotopic estimates from sensor nodes for enhancing estimation performance. The designed DZFEs outperform local zonotopic estimates in terms of estimation performance and strictly preserve the state inclusion

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property. The first DZFE achieves optimal performance under the Frobenius norm via an analytical solution provided in Lemma 2. The second DZFE improves the zonotope enclosure by constructing tight strips for the intersection of local zonotopic estimates. The third DZFE employs a sequential fusing strategy based on the arrival order of local zonotopic estimates, effectively reducing computational burden and communication delay. Stability conditions were derived based on the performance superiority principle, indicating that the boundedness of the fusion estimates depends only on the existence of at least one bounded local estimate. The effectiveness of the proposed methods has been demonstrated through illustrative examples.

Multi-sensor fusion estimation is a field that deeply integrates theoretical and practical aspects. Future work will focus on applying the proposed zonotope fusion criteria to more practical scenarios, such as cooperative localization and path planning.

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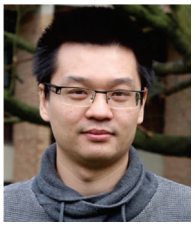
This article has been accepted for publication in IEEE Transactions on Automatic Control. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TAC.2025.3647564



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